

Foundations of Data Science

Lecture 2: Data and exploratory analysis

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Data



What is data?

Data are **recorded observations** about the world.

They can take many forms, including numbers, text, images, and more.

A running example



We will follow one dataset through this lecture: the **Palmer Archipelago penguins**, 344 birds of three species measured on three islands between 2007 and 2009.

Someone stood in the cold, caught a penguin, measured its bill with a caliper and put it on a scale. What follows is about what such records are, and what they are not.

Mathematically, data can be viewed as a function f that maps real-world entities ω to measurable values $\mathbf{x}$,

$$f : \Omega \rightarrow \mathcal{X},$$

where

- Ω is the sample space of possible states ω of the world,
- $\mathcal{X}$ is the measurement space of possible observations $\mathbf{x}$.

Examples:

- Penguin body mass: $\omega \in \{\text{penguins}\} \rightarrow x \in \mathbb{R}^+$ (g)
- Stock price: $\omega \in \{\text{market states}\} \rightarrow x \in \mathbb{R}^+$ (USD)
- Pixel colour: $\omega \in \{\text{scenes}\} \rightarrow \mathbf{x} \in \{0, \dots, 255\}^3$ (RGB values)

If the sample space Ω carries a probability distribution $p(\omega)$, then f turns a random state of the world into a random observation $\mathbf{x} = f(\omega)$. Its distribution is the one f induces from p , called the **data distribution** $p_r(\mathbf{x})$, where r stands for "real".

When $\mathcal{X}$ is continuous, it can be written as

$$p_r(\mathbf{x}) = \int_{\omega \in \Omega} p(\omega) \delta(\mathbf{x} - f(\omega)) d\omega,$$

where δ is the Dirac delta function. When $\mathcal{X}$ is discrete, p_r assigns probabilities rather than a density.

The mechanism that produces the data, here $p(\omega)$ and f , is the **data generating process**. p_r is the distribution it induces.

The measurement process

The measurement process is part of the data generating process. We make it explicit by adding the measurement conditions $\xi \in \Xi$ (instrument settings, environmental conditions, observer effects) to the map,

$$f : \Omega \times \Xi \rightarrow \mathcal{X}.$$

Measurements can introduce quantization (continuous to discrete), noise (random perturbations), and bias (systematic deviations). If $\Omega \times \Xi$ carries a joint distribution $p(\omega, \xi)$, then

$$p_r(\mathbf{x}) = \iint_{\omega \in \Omega, \xi \in \Xi} p(\omega, \xi) \delta(\mathbf{x} - f(\omega, \xi)) d\omega d\xi,$$

which captures the variability of both the phenomenon and its measurement.

Finally, not every entity ends up in the data. Writing $s = 1$ for "this measurement was recorded", what we observe is

$$p_r(\mathbf{x}) = \iint_{\omega \in \Omega, \xi \in \Xi} p(\omega, \xi \mid s = 1) \delta(\mathbf{x} - f(\omega, \xi)) d\omega d\xi.$$

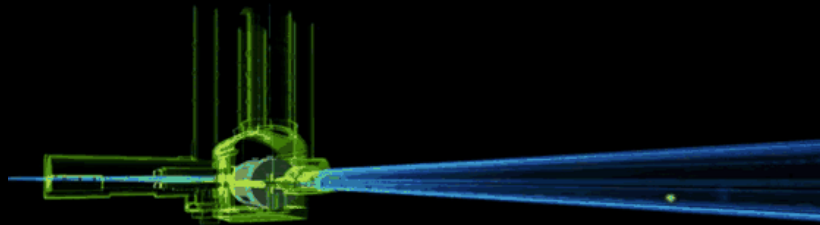
The world enters through $p(\omega)$, the measurement through ξ , and the **selection** through s . When the chance of being recorded depends on ω , p_r describes the entities that were recorded, not the population.



Example 1. A penguin record.

- ω : a penguin of the Palmer Archipelago.
- f : catching it at its nest, reading calipers and a ruler, weighing it in a bag, taking blood for the lab.
- ξ : the season, the observer, each instrument and the resolution it is read at.
- s : only breeding adults, caught at nests holding one egg.

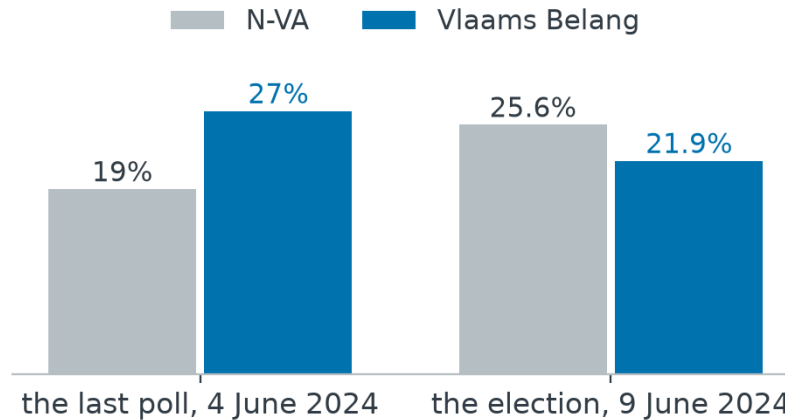
The instruments are visible in the data: body masses are multiples of 25 g, bill lengths have one decimal, flipper lengths are whole millimetres.



Example 2. A collision at the LHC.

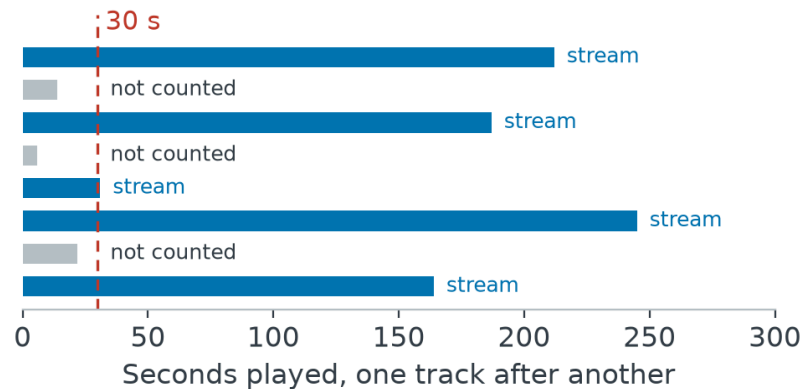
- ω : a bunch crossing, one of 40 million a second.
- f : the detector turning the crossing into electrical signals, and the reconstruction turning those into tracks and energies.
- ξ : the calibration, and the tens of collisions piled up in the same crossing.
- s : the **trigger**, which keeps some 3000 crossings a second and drops the rest.

Votes in Flanders, Chamber of Representatives



Example 3. An opinion in a survey.

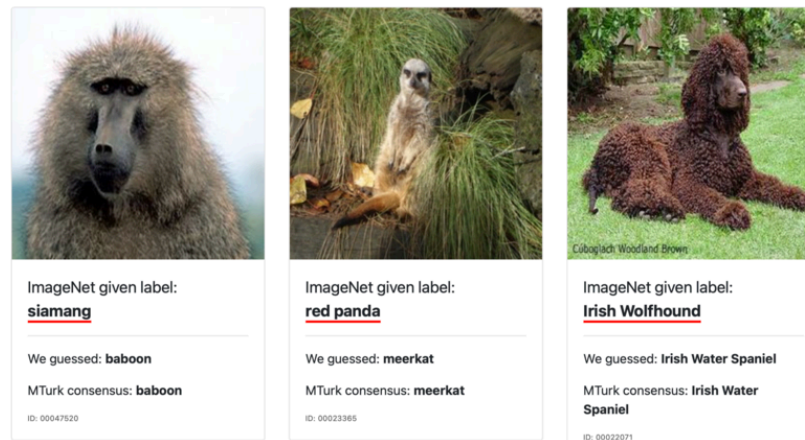
- ω : a voter, whose intention nobody can see.
- f : reaching them, reading a question, writing down the answer.
- ξ : telephone, web or doorstep, the wording and the order of the questions, the day.
- s : who could be contacted, and who agreed to answer.



Example 4. A stream on Spotify.

- ω : a listener, and what they actually like.
- f : logging every play, and counting a **stream** after 30 seconds.
- ξ : the device, the playlist or recommendation that started the track.
- s : plays under 30 seconds are dropped.

These data exist to pay artists, in proportion to streams, and to profile listeners, for recommendations and advertising.



Example 5. A label in a dataset.

- ω : an image, which exists before anyone labels it.
- f : a person looking at the image, applying the guidelines and clicking a category.
- ξ : the guidelines, the annotators and their fatigue, the interface, the rule that settles disagreement.
- s : which images were collected from the web, and which were kept.

Data types

Atomic data are the indivisible units of information collected through measurements. They are often categorized based on their nature and the operations that can be performed on it.

- Numerical (continuous, discrete)
- Categorical (nominal, ordinal)

Numerical data

- Continuous: $x \in \mathbb{R}$ (e.g., temperature), $x \in \mathbb{R}^+$ (e.g., height, weight)
- Discrete: $x \in \mathbb{N}$ (e.g., counts), $x \in \mathbb{Z}$ (e.g., differences)

Categorical data

- Nominal: $x \in \mathcal{C} = \{c_1, c_2, \dots, c_K\}$ (e.g., colors, types, text characters) without intrinsic order
- Ordinal: $x \in \mathcal{C}$ with ordering relations $c_1 \prec c_2 \prec \dots \prec c_K$ (e.g., ratings, grades)



Example: one penguin record

```
species: Adelie           # Categorical, nominal
island: Torgersen         # Categorical, nominal
bill length: 39.1 mm      # Numerical, continuous
body mass: 3750 g         # Numerical, continuous
sex: male                 # Categorical, nominal
year: 2007                # Numerical, discrete
```

No variable here is ordinal; a rating from 1 to 5, or a grade, would be.

Data structures

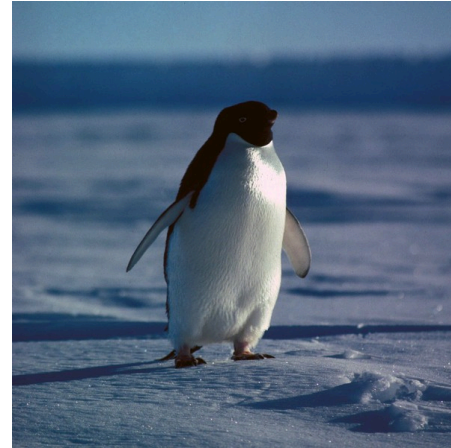
A measurement is either a single atomic value x , or a composite $\mathbf{x}$ made of several of them. Common composite structures include:

- Tabular data
- Arrays and tensors
- Sequences
- Networks and graphs

Tabular data

	species	island	bill length (mm)	body mass (g)	sex
0	Adelie	Torgersen	39.1	3750.0	male
1	Adelie	Torgersen	39.5	3800.0	female
2	Adelie	Torgersen	40.3	3250.0	female
3	Adelie	Torgersen	NaN	NaN	NaN
4	Adelie	Torgersen	36.7	3450.0	female
...	...	...	...	...	...
339	Chinstrap	Dream	55.8	4000.0	male
340	Chinstrap	Dream	43.5	3400.0	female
341	Chinstrap	Dream	49.6	3775.0	male
342	Chinstrap	Dream	50.8	4100.0	male
343	Chinstrap	Dream	50.2	3775.0	female

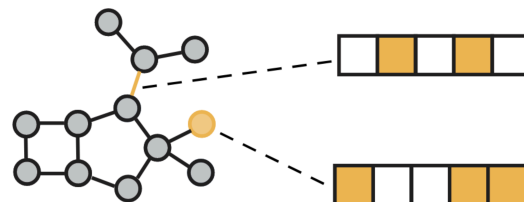
Arrays and tensors



Sequences

```
['P', 'e', 'n', 'g', 'u', 'i',  
 'n']
```

Networks and graphs



A data frame $\mathbf{X}$ represents a **tabular collection** of N records (rows) over d variables/atomic measurements (columns),

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{pmatrix}.$$

Each entry x_{ij} corresponds to the value of variable j for record i .

Variables are often heterogeneous (mixing numerical and categorical types). When all variables are numerical, the data frame can be viewed as a matrix $\mathbf{X} \in \mathbb{R}^{N \times d}$.

For the penguins, $N = 344$ records over $d = 8$ variables, four of them numerical.

Notation

Bold marks an object holding more than one number. Plain marks a single one.

- $\mathbf{x}$ is a measurement, $\mathcal{X}$ the space it lives in.
- $\mathbf{X} \in \mathbb{R}^{N \times d}$ is the data frame, with N records indexed by i and d variables indexed by j .
- $\mathbf{x}_i$ is a record, the i -th row of $\mathbf{X}$; x_{ij} is a single entry.

Later in the lecture, $\mathbf{x}_j$ denotes the column of variable j : the index letter says whether a row or a column is meant.

Values of the same kind, in the same unit, can be stacked into **arrays** or **tensors** $\mathbf{x} \in \mathbb{R}^{d_1 \times d_2 \times \dots \times d_k}$, where the position of a value stands for a place or a time.

- Images: 3d arrays $\mathbf{x} \in \{0, \dots, 255\}^{h \times w \times c}$ (height, width, channels). The photo on the previous slide is such an array.
- Videos: 4d arrays $\mathbf{x} \in \{0, \dots, 255\}^{t \times h \times w \times c}$ (time, height, width, channels).

Data can also be structured as ordered **sequences** $\mathbf{x} = (x_1, x_2, \dots, x_T)$ indexed by time or position. Each element is atomic here, but may itself be composite.

- Time series: x_t is a measurement at time t (e.g., a stock price).
- Text: x_t is the t -th word in a document.

Finally, data can be organized as **networks** or **graphs** $\mathbf{x} = (V, E)$, where entities are represented as nodes V and relationships as edges E . Nodes and edges may carry their own attributes, $\mathbf{x}_v$ and $\mathbf{x}_{uv}$.

Examples:

- Molecular structures, where nodes represent atoms and edges represent bonds.
- Social networks, where nodes represent individuals and edges represent interactions or relationships.



Follow the tutorials in `nb02a-tables.ipynb` , `nb02b-jax.ipynb` , and `nb02c-data-wrangling.ipynb` to practice working with arrays and data frames in Python.

Data quality

Real-world data are imperfect, and each **quality issue** sits somewhere in the process that produced them:

- **Missing values:** nothing was recorded, for a whole record, which is $s = 0$, or for one of its entries.
- **Measurement errors:** $f(\omega, \xi)$ is off the quantity of interest, through the noise and the bias that ξ brings.
- **Outliers:** a value far from the others, because ξ went wrong or because ω was rare.

Missing values are common in real-world datasets and can arise from various factors such as non-response in surveys, sensor malfunctions, or data corruption.

Let $\mathbf{X}_{\text{full}}$ be the complete data and $\mathbf{M} \in \{0, 1\}^{N \times d}$ the missingness pattern, with $m_{ij} = 1$ when entry ij is observed and $m_{ij} = 0$ when it is missing¹.

What we actually hold is the pair $(\mathbf{X}_{\text{obs}}, \mathbf{M})$, where $\mathbf{X}_{\text{obs}} = \{x_{ij} : m_{ij} = 1\}$ are the observed entries and $\mathbf{X}_{\text{mis}} = \{x_{ij} : m_{ij} = 0\}$ the missing ones.

1: Rubin's convention is the opposite, 1 for missing.

	species	island	bill len.	bill depth	flipper len.	body mass	sex
3	Adelie	Torgersen	NaN	NaN	NaN	NaN	NaN
8	Adelie	Torgersen	34.1	18.1	193.0	3475.0	NaN
9	Adelie	Torgersen	42.0	20.2	190.0	4250.0	NaN
10	Adelie	Torgersen	37.8	17.1	186.0	3300.0	NaN
11	Adelie	Torgersen	37.8	17.3	180.0	3700.0	NaN
47	Adelie	Dream	37.5	18.9	179.0	2975.0	NaN
178	Gentoo	Biscoe	44.5	14.3	216.0	4100.0	NaN
218	Gentoo	Biscoe	46.2	14.4	214.0	4650.0	NaN
256	Gentoo	Biscoe	47.3	13.8	216.0	4725.0	NaN
268	Gentoo	Biscoe	44.5	15.7	217.0	4875.0	NaN
271	Gentoo	Biscoe	NaN	NaN	NaN	NaN	NaN

The 11 penguin records with missing entries: two birds were never measured, nine have no recorded sex.

The pattern $\mathbf{M}$ is itself produced by the measurement process, and is modeled as such:

- Missing completely at random (MCAR): the pattern is independent of the data, $p(\mathbf{M} \mid \mathbf{X}_{\text{full}}) = p(\mathbf{M})$.
- Missing at random (MAR): the pattern may depend on what is observed, but not on what is missing, $p(\mathbf{M} \mid \mathbf{X}_{\text{full}}) = p(\mathbf{M} \mid \mathbf{X}_{\text{obs}})$.
- Missing not at random (MNAR): the pattern depends on the missing entries themselves.

What we hold is the pair $(\mathbf{X}_{\text{obs}}, \mathbf{M})$, whose joint law is

$$p(\mathbf{X}_{\text{obs}}, \mathbf{M}) = \int p(\mathbf{X}_{\text{full}}) p(\mathbf{M} \mid \mathbf{X}_{\text{full}}) d\mathbf{X}_{\text{mis}}.$$

Under MAR, $p(\mathbf{M} \mid \mathbf{X}_{\text{full}}) = p(\mathbf{M} \mid \mathbf{X}_{\text{obs}})$ does not depend on $\mathbf{X}_{\text{mis}}$, so it leaves the integral,

$$p(\mathbf{X}_{\text{obs}}, \mathbf{M}) = p(\mathbf{M} \mid \mathbf{X}_{\text{obs}}) \int p(\mathbf{X}_{\text{full}}) d\mathbf{X}_{\text{mis}} = p(\mathbf{M} \mid \mathbf{X}_{\text{obs}}) p(\mathbf{X}_{\text{obs}}).$$

If the two factors share no parameters, fitting a model to $\mathbf{X}_{\text{obs}}$ alone is enough. Under MNAR the factor stays inside the integral, and the mechanism has to be modelled together with the data.

Example: a survey of $N = 1000$ respondents, where 30% do not answer the income question.

- MCAR: Randomly selected respondents skip the question.
- MAR: Younger respondents are less likely to answer.
- MNAR: High earners refuse to answer.

Imputing or discarding missing values without making explicit the assumptions about the missingness mechanism can lead to biased results.

An **outlier** is a value far from the rest. It can come from either end of the measurement process:

- from ξ : an instrument that failed, a value typed in the wrong column, a unit mistaken;
- from ω : an entity that is genuinely unusual, measured correctly.

The value alone does not say which.

Two models account for outliers explicitly.

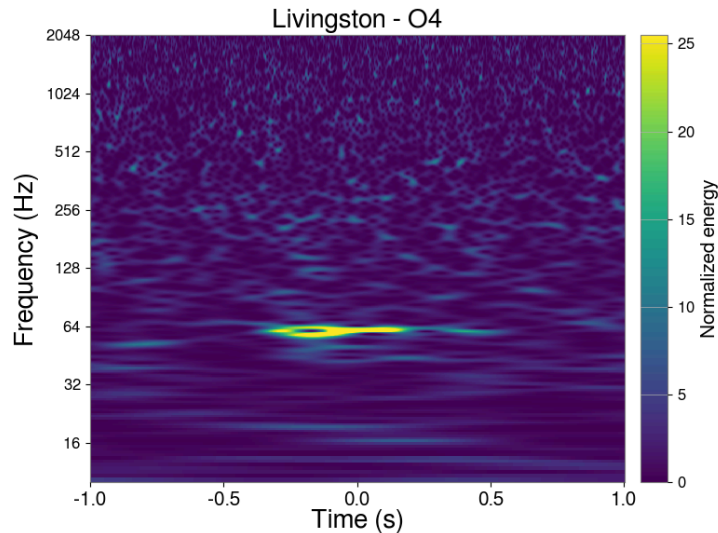
A **contamination mixture** gives bad measurements their own distribution,

$$p(\mathbf{x}) = (1 - \varepsilon)p_{\text{model}}(\mathbf{x}) + \varepsilon p_{\text{bad}}(\mathbf{x}),$$

where ε is the proportion of bad measurements.

A **heavy-tailed** data model, a Student's t in place of a Gaussian, assigns a larger probability to large deviations.

Outliers should not be removed blindly unless explicitly justified by the measurement model or domain knowledge.



Example 1. Glitches in gravitational wave detectors are outliers that can mimic true signals and complicate detection efforts. They can arise from environmental disturbances, instrumental artifacts, or other non-astrophysical sources.



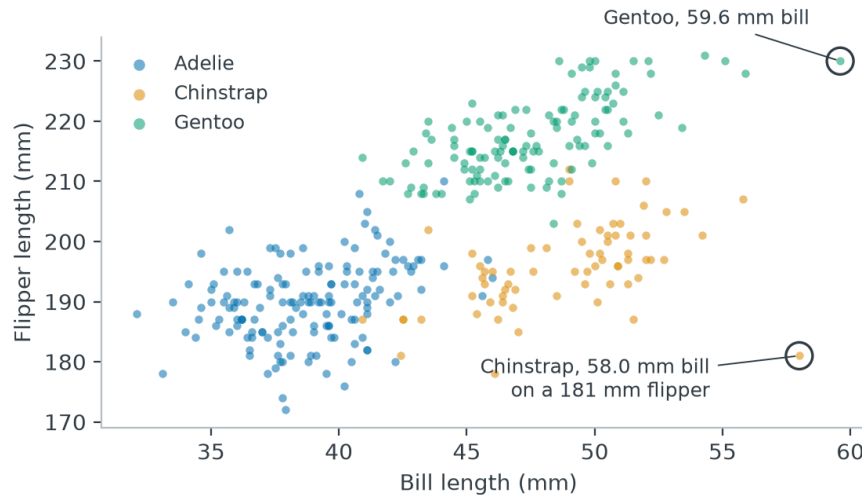
Example 2. Temperature readings in Liège.

20.1, 19.8, 20.3, 1000.0, 20.2, 19.9

The 1000.0 value is an outlier likely due to a sensor error.

19.2, 19.8, 20.3, 37.8, 20.2, 19.9

The 37.8 value is a rare but plausible measurement on a hot day.



Example 3. Outliers among the penguins.

- Pooled, no measurement is an outlier: the three species hide each other.
- Among the Gentoo, a 59.6 mm bill lies about 4 standard deviations above the mean.
- Among the Chinstrap, one bird is unusual only jointly: neither its 58.0 mm bill nor its 181 mm flipper is extreme on its own.

Where this leaves us

Everything in this part is one picture of how the data came to be:

- the map f from states of the world to measurements,
- the conditions ξ under which measurements are taken,
- the selection s of which entities are recorded at all,
- the pattern $\mathbf{M}$ of which entries are missing,
- the data distribution $p_r(\mathbf{x})$ that all of this induces.

Exploratory data analysis

Exploratory data analysis (EDA) is the systematic examination of data to understand its structure, patterns and anomalies before formal modeling. It typically involves visual and quantitative techniques to summarize key characteristics of the data.

The goal is to generate hypotheses and inform modeling decisions, not to confirm preconceived notions.

What we are looking at

All we hold is N records, the rows $\mathbf{x}_i = (x_{i1}, \dots, x_{id})$ of the data frame. Their **empirical distribution** puts equal mass on each of them,

$$\hat{p}_N(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N \delta(\mathbf{x} - \mathbf{x}_i).$$

$\hat{p}_N$ is not p_r . It is what N records show of it, and that gap is what the rest of the course is about.



Back to the penguins, with the whole table in hand: 344 records, four numerical measurements, three species and three islands.

Switch to `nb02d-eda.ipynb` to follow along.

Univariate analysis

Let us consider a variable j of the data frame. We write x_j for that variable as a random quantity, distributed under p_r , and $\mathbf{x}_j = (x_{1j}, x_{2j}, \dots, x_{Nj})^T$ for the column of values actually recorded.

Univariate analysis focuses on understanding the distribution and characteristics of this single variable.

Look at the values first: print the column, scroll through it.

Then, for numerical variables:

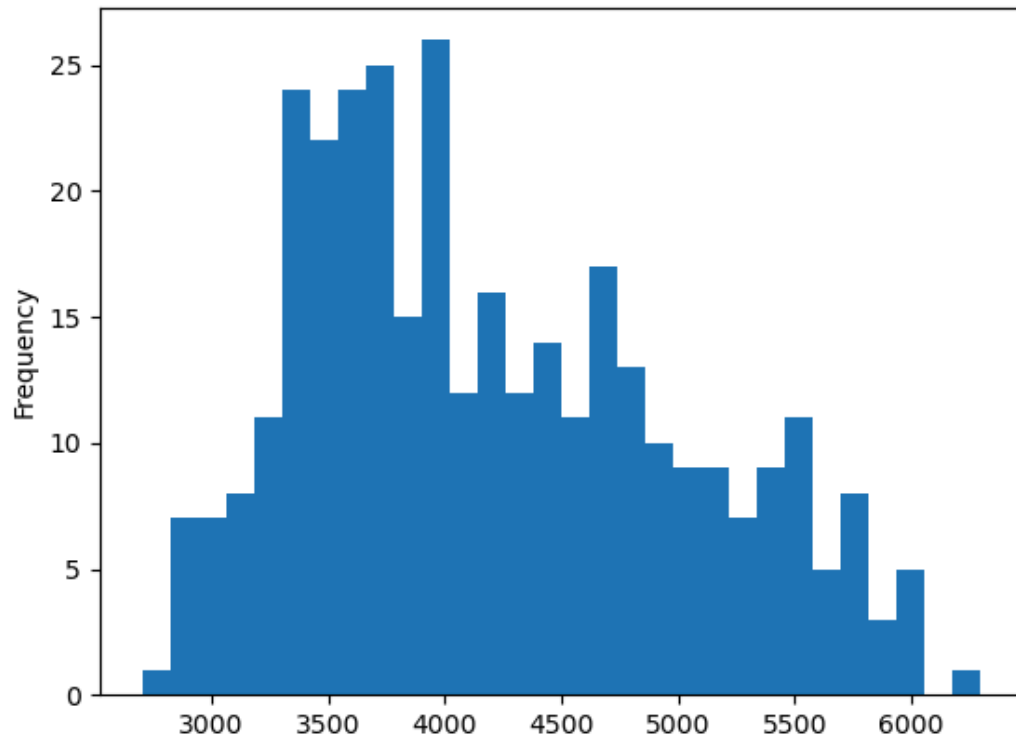
- Plotting the data: histograms reveal the distribution shape.
- Summarizing with statistics: mean, median, mode, variance, skewness, kurtosis.

For categorical variables:

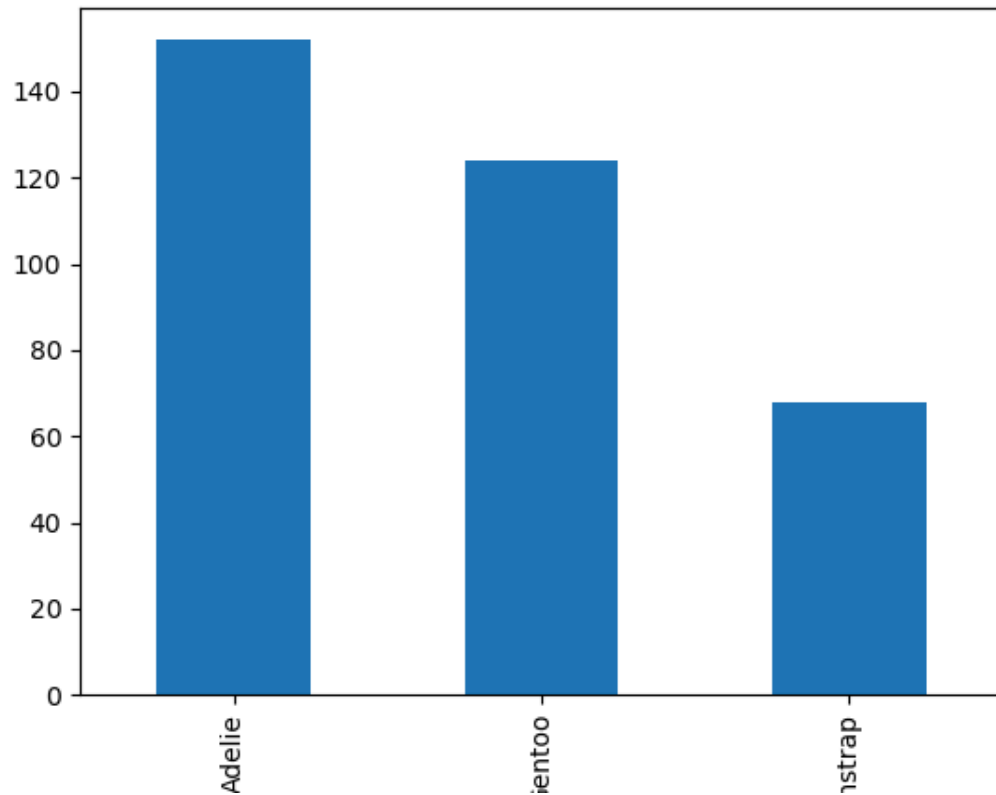
- Counting occurrences of each category.

3750	3800	3250	NA	3450	3650	3625	4675	3475	4250	3300	3700	3200	3800
4400	3700	3450	4500	3325	4200	3400	3600	3800	3950	3800	3800	3550	3200
3150	3950	3250	3900	3300	3900	3325	4150	3950	3550	3300	4650	3150	3900
3100	4400	3000	4600	3425	2975	3450	4150	3500	4300	3450	4050	2900	3700
3550	3800	2850	3750	3150	4400	3600	4050	2850	3950	3350	4100	3050	4450
3600	3900	3550	4150	3700	4250	3700	3900	3550	4000	3200	4700	3800	4200
3350	3550	3800	3500	3950	3600	3550	4300	3400	4450	3300	4300	3700	4350
2900	4100	3725	4725	3075	4250	2925	3550	3750	3900	3175	4775	3825	4600
3200	4275	3900	4075	2900	3775	3350	3325	3150	3500	3450	3875	3050	4000
3275	4300	3050	4000	3325	3500	3500	4475	3425	3900	3175	3975	3400	4250
3400	3475	3050	3725	3000	3650	4250	3475	3450	3750	3700	4000	4500	5700
4450	5700	5400	4550	4800	5200	4400	5150	4650	5550	4650	5850	4200	5850
4150	6300	4800	5350	5700	5000	4400	5050	5000	5100	4100	5650	4600	5550
5250	4700	5050	6050	5150	5400	4950	5250	4350	5350	3950	5700	4300	4750
5550	4900	4200	5400	5100	5300	4850	5300	4400	5000	4900	5050	4300	5000
4450	5550	4200	5300	4400	5650	4700	5700	4650	5800	4700	5550	4750	5000
5100	5200	4700	5800	4600	6000	4750	5950	4625	5450	4725	5350	4750	5600
4600	5300	4875	5550	4950	5400	4750	5650	4850	5200	4925	4875	4625	5250
4850	5600	4975	5500	4725	5500	4700	5500	4575	5500	5000	5950	4650	5500
4375	5850	4875	6000	4925	NA	4850	5750	5200	5400	3500	3900	3650	3525
3725	3950	3250	3750	4150	3700	3800	3775	3700	4050	3575	4050	3300	3700
3450	4400	3600	3400	2900	3800	3300	4150	3400	3800	3700	4550	3200	4300
3350	4100	3600	3900	3850	4800	2700	4500	3950	3650	3550	3500	3675	4450
3400	4300	3250	3675	3325	3950	3600	4050	3350	3450	3250	4050	3800	3525
3950	3650	3650	4000	3400	3775	4100	3775						

The 344 body masses, in the order they were recorded.



Numerical: Histogram of body mass for all penguins.
One broad mode with a long right tail: perhaps not a single population.



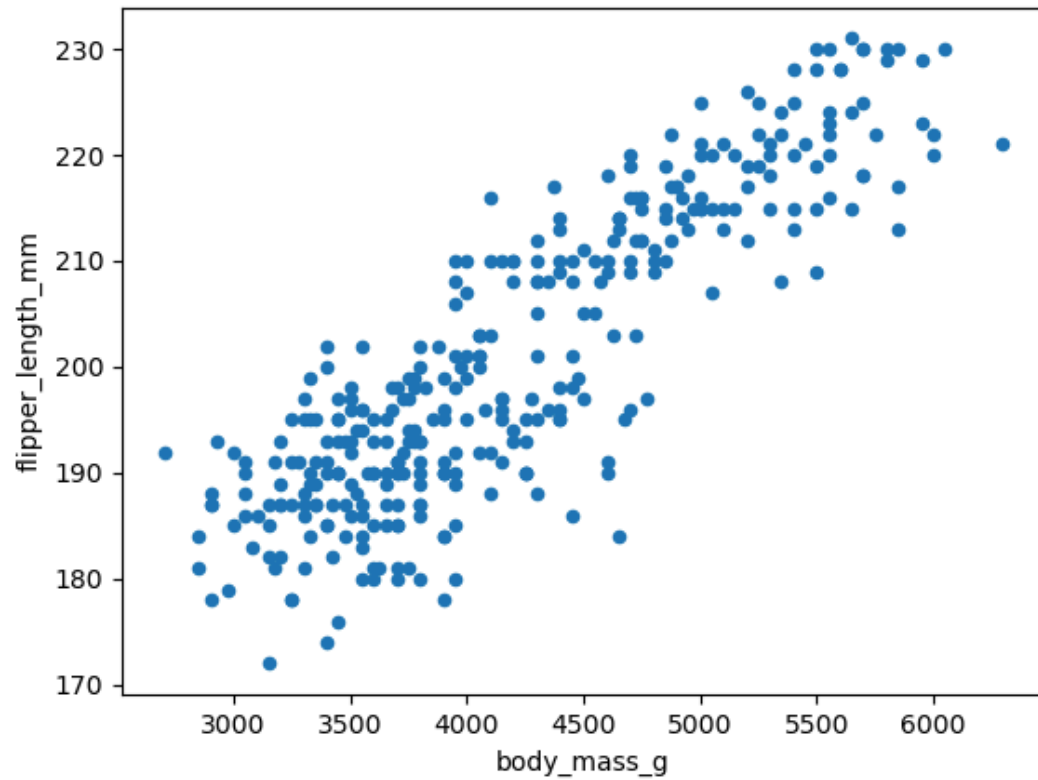
Categorical: Bar plot of species counts for all penguins.
Unbalanced groups: 152 Adelie, 124 Gentoo, 68 Chinstrap.

Bivariate analysis

Bivariate analysis examines the relationship between two variables j and k , from their recorded columns $\mathbf{x}_j$ and $\mathbf{x}_k$.

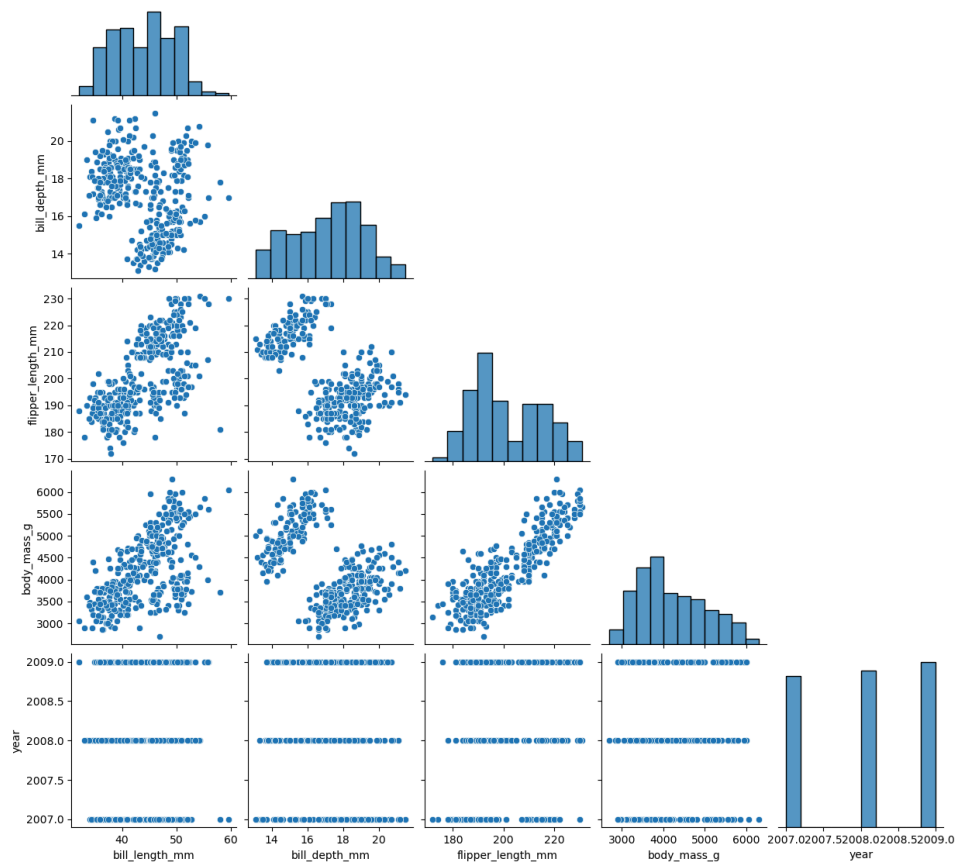
Depending on the types of variables, different techniques are used:

- Pair plots for two numerical variables (scatter or 2d histogram).
- Multiple histograms for categorical vs numerical variables.
- Contingency tables for two categorical variables.
- Correlation coefficients (e.g., Pearson, Spearman) for numerical variables.



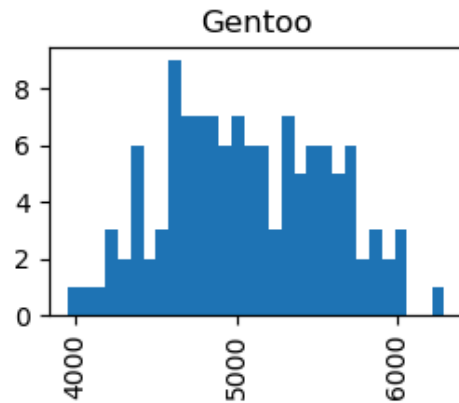
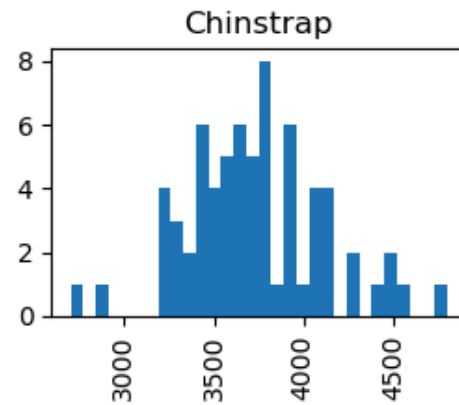
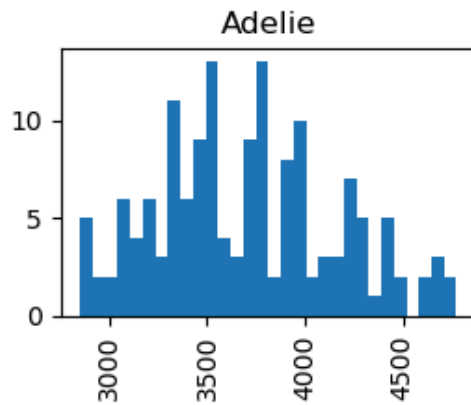
Numerical vs. numerical: Scatter plot of body mass vs flipper length.

Mass grows with flipper length, close to linearly.



Pair plots of all numerical variables.

Most panels show two or three clouds rather than one.



Categorical vs. numerical: Histograms of body mass by species.

*The right tail was Gentoo, at 5076 g on average
against about 3700 g for the others.*

island	Biscoe	Dream	Torgersen
species			
Adelie	44	56	52
Chinstrap	0	68	0
Gentoo	124	0	0

Categorical vs. categorical: Contingency table of species and island.

*Gentoo live only on Biscoe, Chinstrap only on Dream:
species and island are dependent.*

Correlation coefficients can quantify the dependency between two numerical variables. They are useful but come with assumptions and limitations.

- Pearson correlation measures linear relationships. Its empirical version reads

$$\hat{\rho}_{jk} = \frac{\sum_{i=1}^N (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)}{\sqrt{\sum_{i=1}^N (x_{ij} - \bar{x}_j)^2} \sqrt{\sum_{i=1}^N (x_{ik} - \bar{x}_k)^2}},$$

an estimate of $\rho_{jk} = \text{cov}(x_j, x_k) / (\sigma_j \sigma_k)$ under p_r . It ignores non-linear dependencies.

- Spearman correlation is the Pearson correlation of the rank-transformed variables. It captures monotonic relationships.
- Correlation does not imply causation and can be affected by outliers.

The **mutual information** between two variables x_j and x_k measures the reduction in uncertainty about one given knowledge of the other. For discrete variables,

$$I(x_j; x_k) = \sum_{x_j} \sum_{x_k} p_r(x_j, x_k) \log \frac{p_r(x_j, x_k)}{p_r(x_j)p_r(x_k)},$$

with integrals in place of the sums for continuous ones.

Mutual information captures any statistical relationship, not just linear or monotonic ones. It is a property of p_r , however, which must itself be estimated from the N records, which can be challenging.

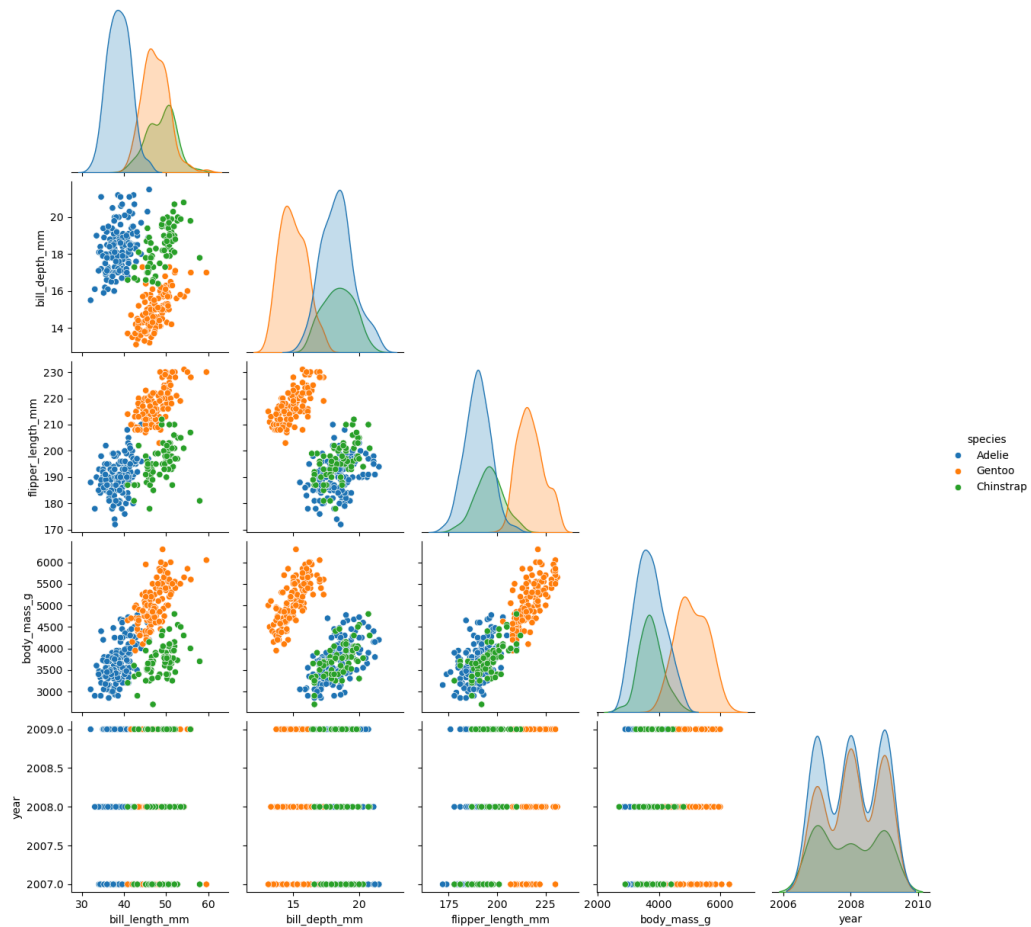
Multivariate analysis

Multivariate analysis explores relationships among three or more variables at once.

Common techniques include:

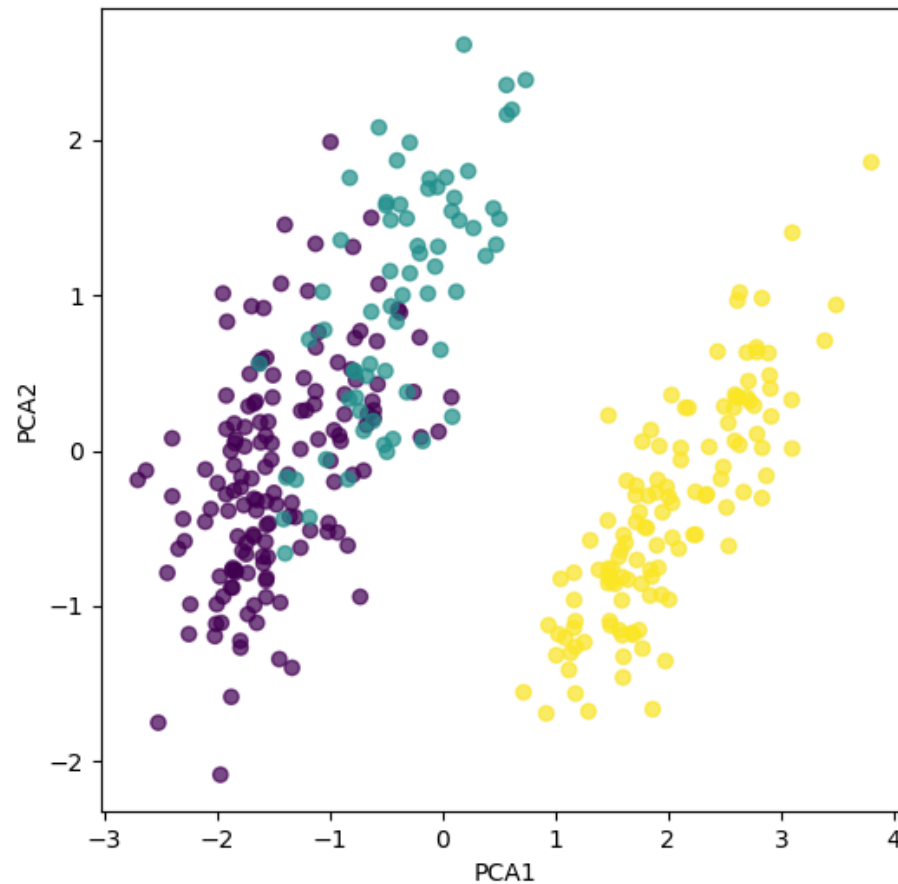
- The same as bivariate analysis, but conditioning on a third variable (e.g., pair plots colored by species).
- Dimensionality reduction methods (e.g., PCA, t-SNE) to visualize high-dimensional data.
- Clustering algorithms (e.g., k-means, hierarchical clustering) to identify groups of similar records.

The last two work on the numerical columns of **X**.



Pair plots of all numerical variables, colored by species.

Colour by species and the clouds line up.



PCA projection of all numerical variables, colored by species.
Two components are enough to separate the species.

From observations to hypotheses

The analysis of the penguins suggests four models:

- Body mass is not a single population, since the sample mixes three species (mixture models, Lecture 4).
- Body mass increases with flipper length (regression, Lecture 6).
- Species and island are dependent, and measurements differ from group to group (hierarchical models, Lecture 4).
- The four measurements are strongly correlated, so fewer dimensions may be enough to describe them (latent variable models, Lecture 4).

The plots establish none of this. They suggest models, which then have to be built, fitted and criticized.

EDA does not stop once a model is built. In the critique step, the same plots come back on what the model gets wrong.

- Residuals reveal patterns the model does not capture.
- Prediction errors point to subsets of the data that are problematic.
- Unexpected patterns suggest new variables or a revised model.

Next lecture: how to draw all of these plots well.

